Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 6

Roll No. : 496

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01) WAP to find sum of all the elements in a List.

In [1]:
```
length = int(input("Enter number of elements: "))
numbers = []

for i in range(length):
    num = int(input(f"Enter element {i+1}: "))
    numbers.append(num)

total = sum(numbers)
```

```
print("List elements:", numbers)
print("Sum of all elements:", total)
```

```
List elements: [1, 2, 3, 4, 5]
Sum of all elements: 15
```

02) WAP to find largest element in a List.

In [2]:
```
length = int(input("Enter number of elements: "))
numbers = []

for i in range(length):
    num = int(input(f"Enter element {i+1}: "))
    numbers.append(num)

max = max(numbers)

print("List elements :" , numbers)
print("Maximum element :" ,max )
```

```
List elements : [3, 1, 4, 5, 2]
Maximum element : 5
```

03) WAP to find the length of a List.

In [4]:
```
numbers = [1,2,3,4,5,6,7]
print("List elements :" , numbers)
print("length of List :" , len(numbers))
```

```
List elements : [1, 2, 3, 4, 5, 6, 7]
length of List : 7
```

04) WAP to interchange first and last elements in a list.

In [12]:
```
length = int(input("Enter number of elements: "))
numbers = []

for i in range(length):
    num = int(input(f"Enter element {i+1}: "))
```

```
    numbers.append(num)

print("List elements :" , numbers)

numbers[0] , numbers[1] = numbers[1] , numbers[0]

print(" New List elements :" , numbers)

# numbers = [1,2,3,4,5,6,7]
# length = len(numbers)

# print("List elements :" , numbers)

# temp = numbers[0]
# numbers[0] = numbers[length-1]
# numbers[length-1] = temp

# print(" New List elements :" , numbers)
```

```
List elements : [2, 1, 3, 4, 5]
 New List elements : [1, 2, 3, 4, 5]
```

05) WAP to split the List into two parts and append the first part to the end.

In [10]:

```
length = int(input("Enter number of elements: "))
numbers = []

for i in range(length):
    numbers.append(int(input(f"Enter element {i+1}: ")))

position = int(input("Enter position to split the list: "))

result = numbers[position:] + numbers[:position]

print("Original List:", numbers)
print("Modified List:", result)
```

```
Original List: [1, 2, 3, 4]
Modified List: [4, 1, 2, 3]
```

06) WAP to interchange the elements on two positions entered by a user.

In [13]:
```
length = int(input("Enter number of elements: "))
numbers = []

for i in range(length):
    numbers.append(int(input(f"Enter element {i+1}: ")))

position1 , position2 = input("Enter position of elements: ").split(" ")
position1 = int(position1) - 1
position2 = int(position2) -1
print("list elements :" , numbers)

numbers[position1] , numbers[position2] = numbers[position2] , numbers[position1]

print("New list elements :" , numbers)
```

```
list elements : [5, 2, 3, 4, 1]
New list elements : [1, 2, 3, 4, 5]
```

07) WAP to reverse the list entered by user.

In [15]:
```
length = int(input("Enter number of elements: "))
numbers = []

for i in range(length):
    numbers.append(int(input(f"Enter element {i+1}: ")))

print("Original List:", numbers)
numbers.reverse()
print("Reversed List:", numbers)
```

```
Original List: [1, 2, 3, 4, 5]
Reversed List: [5, 4, 3, 2, 1]
```

08) WAP to print even numbers in a list.

In [26]:
```
numbers = list(map(int , input("Enter numbers with space:").split(" ")))
result = [i for i in numbers if i%2==0]
print("Original List:", numbers)
print("Even numbers :",result)
```

```
Original List: [1, 2, 3, 4, 5]
Even numbers : [2, 4]
```

09) WAP to count unique items in a list.

In [28]:
```
numbers = list(map(int , input("Enter numbers with space:").split(" ")))
count1=0
for i in numbers:
    if numbers.count(i) == 1:
        count1+=1
print("Original List:", numbers)
print("total Unique items:", count1)
```

```
Original List: [1, 1, 2, 3, 4, 5, 6, 6]
total Unique items: 4
```

10) WAP to copy a list.

In [30]:
```
import copy
numbers = list(map(int , input("Enter numbers with space:").split(" ")))
numbercopy = numbers.copy()

print("Original List:", numbers)
print("Copied List:", numbercopy)
```

```
Original List: [1, 2, 3, 4, 5]
Copied List: [1, 2, 3, 4, 5]
```

11) WAP to print all odd numbers in a given range.

In [36]:
```
start , end = input("Enter Starting and Ending Numbers :").split()
start = int(start)
start = start+1 if start%2==0 else start
```

```
end = int(end)

numbers= [ i for i in range(start ,end ,2)]
print("Odd numbers :",numbers)
```

```
Odd numbers : [3, 5, 7, 9]
```

12) WAP to count occurrences of an element in a list.

In [56]:
```
numbers = list(map(int , input("Enter numbers with space:").split(" ")))
count = []
for i in numbers:
    if i not in count:
        print("Count of",i,"=",numbers.count(i))
        count.append(i)
```

```
Count of 1 = 3
Count of 2 = 2
Count of 3 = 1
Count of 4 = 3
Count of 5 = 1
```

13) WAP to find second largest number in a list.

In [44]:
```
numbers = list(map(int , input("Enter numbers with space:").split(" ")))
numbers.sort()
max1 = numbers[-2]
print("List elements :" , numbers)
print("second larget element :" ,max1 )
```

```
List elements : [1, 2, 3, 4, 5]
second larget element : 4
```

14) WAP to extract elements with frequency greater than K.

In [54]:
```
numbers = list(map(int , input("Enter numbers with space:").split(" ")))
k= int(input("Enter Frequency :"))
res = []
for i in numbers:
```

```
    if numbers.count(i) > k:
            res.append(i)

print("Original List:", numbers)
print("frequency greater than K items:",list(set(res)))
```

```
Original List: [1, 1, 1, 2, 2, 3, 4, 4, 4, 5]
frequency greater than K items: [1, 4]
```

15) WAP to create a list of squared numbers from 0 to 9 with and without using List Comprehension.

In [57]:
```
res=[]
for i in range(1,10):
        res.append(i**2)
print(res)
```

```
[1, 4, 9, 16, 25, 36, 49, 64, 81]
```

16) WAP to create a new list (fruit whose name starts with 'b') from the list of fruits given by user.

In [67]:
```
fruits = input("Enter numbers with space:").split(" ")
res=[]
for i in fruits:
    if i[0]  in "bB":
        res.append(i)
print(res)
```

```
['banana', 'blackberry', 'Blueberry']
```

17) WAP to create a list of common elements from given two lists.

In [1]:
```
numbers1 = list(map(int , input("Enter numbers with space:").split(" ")))
numbers2 = list(map(int , input("Enter numbers with space:").split(" ")))
common = [i for i in numbers1 if i in numbers2]

print(common)
```

```
[4, 5, 6]
```

In [ ]:

```

```